Cell-Level Shower Shape Corrections for Photon ID in MC23e

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LIP / ATLAS

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ATLAS Internal

Outline

1. Motivation & Context
2. Shower Shape Variables
3. Cell-Energy Reweighting Method
4. Closure Test
5. Results
6. Conclusions & Next Steps

Motivation

Photon identification in ATLAS relies on rectangular cuts on shower shape variables, computed from cell energies in the EM calorimeter.

Problem: well-known mismodelling of calorimeter shower shapes in the Geant4 simulation.

Several sources of mismodelling:

- Detector geometry description in simulation
- Cross-talk — signal leakage between LAr cells due to non-ideal connections
- Magnetic field modelling

- Need to correct these effects to better represent data
- Improving G4 is very time-consuming
- Currently, direct data-driven corrections to the shower shapes are derived, the so-called **fudge factors** (FF)

Current Corrections & Motivation for Cell-Level Approach

Current approach: fudge factors

- Data–MC differences in shower shapes corrected by per-variable **shifts and smearings**
- Derived from $Z \to ee$ control samples, applied to MC
- Each variable corrected *independently*

Limitations:

- Does **not** preserve correlations between variables
- Corrections derived at distribution level, not event level
- Not suitable for ML-based ID (CNN/GNN) that uses cell-level inputs directly
- Must be re-derived for every new variable

Cell-level approach:

- Correct individual **cell energies** in the EM calorimeter cluster
- Recompute *all* shower shapes from corrected cells
- Inter-variable correlations preserved **by construction**
- Any new variable automatically benefits
- Enables consistent input for both cut-based and ML-based ID

Pioneered for Run 2 in ATL-COM-PHYS-2021-640 (F. Caetano et al.) — applied to Layer 2 cells using $Z \to \ell\ell\gamma$ matched data–MC ntuples.

Project Goal

Objective

Continue and extend the cell-level shower shape correction studies for **Run 3 conditions**, building on the Run 2 method developed in ATL-COM-PHYS-2021-640.

Previous work (Run 2, F. Caetano et al.):

- Cell-energy reweighting applied to **Layer 2** of the EM calorimeter
- Used matched data–MC event pairs from $Z \to \ell\ell\gamma$
- Two correction methods studied: flat shift and energy-dependent (TProfile)
- Demonstrated improved data–MC agreement for R_η, R_ϕ, $w_{\eta 2}$

This work — scope:

1. **Validate** the cell-energy reweighting method with Run 3 MC via closure tests
2. **Apply** corrections to real Run 3 data
3. **Extend** the approach to **Layers 1 and 3** (strip and back layers)
4. **Explore normalising flows** as an alternative correction method — learn the full multivariate data–MC transformation at cell level, capturing inter-cell correlations and conditioning on p_{T}, η, conversion type

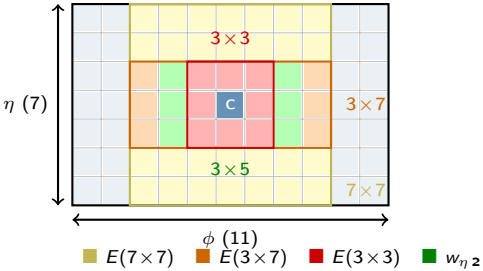
Shower Shape Variables

Three key discriminants:

$$R_\eta = \frac{E(3\times7)}{E(7\times7)}$$

$$R_\phi = \frac{E(3\times3)}{E(3\times7)}$$

$$w_{\eta\,2} = \sqrt{\frac{\sum E_i\,\eta_i^2}{\sum E_i} - \left(\frac{\sum E_i\,\eta_i}{\sum E_i}\right)^2}$$



$E(m\times n)$: energy in $m\,(\eta)\times n\,(\phi)$ sub-window.
$w_{\eta\,2}$ computed in 3×5 window.

Cell-Energy Reweighting: Overview

1. **Compute cell energy fractions** from the 7×11 cluster:

$$f_k = \frac{e_k}{E_{\text{tot}}} = \frac{e_k}{\sum_{j=1}^{77} e_j} \qquad (k = 1, \ldots, 77)$$

2. **Derive correction** by comparing data and MC fractions cell-by-cell, in each of 14 $|\eta|$ bins
3. **Apply correction** to MC fractions: $f_k^{\text{corr}} = f_k^{\text{MC}} + \text{correction}_k$
4. **Reconstruct cell energies** with energy conservation:

$$e_k^{\text{corr}} = \frac{f_k^{\text{corr}}}{\sum_j f_j^{\text{corr}}} \times E_{\text{tot}}$$

5. **Recompute all shower shapes** from the corrected cell energies

Two methods differ in step 2: how the correction is parameterised

Correction Methods

Method 1: Flat Shift

One number per cell per $|\eta|$ bin:

$$\Delta_k = \langle f_k^{\text{data}} \rangle - \langle f_k^{\text{MC}} \rangle$$

Applied as:

$$f_k^{\text{corr}} = f_k^{\text{MC}} + \Delta_k$$

- Simple, robust
- Ignores energy dependence of correction

Method 2: TProfile (Energy-Dependent)

Correction varies with cell fraction:

$$\alpha_k(f) = \left\langle f_k^{\text{data}} - f_k^{\text{MC}} \right\rangle \Big|_{f_k^{\text{MC}} = f}$$

Applied via interpolation:

$$f_k^{\text{corr}} = f_k^{\text{MC}} + \alpha_k(f_k^{\text{MC}})$$

- Captures non-linear effects
- 1078 profile histograms
  (77 cells $\times$ 14 $|\eta|$ bins)

Following ATL-COM-PHYS-2021-640, Sections 5.1 and 5.2 (F. Caetano et al.)

Closure Test: Setup & Inputs

Goal: validate the correction procedure in a controlled environment where the "truth" is known by construction.

Monte Carlo sample

- MC23e $Z \to ee\gamma$, DSID 700770
- Sherpa 2.2.14, Run 3 simulation
- $\sim$9.6 M events total
- $\sim$7.8 M pass quality selection (81%)

Ntuple production

- NTupleMaker package (Athena 25.0.40)
- Layer 2 cell energies stored as vector of 77 doubles (7×11 cluster)
- Cluster barycentre η_2 for binning

$|\eta|$ **binning** — 14 bins, standard ATLAS EM calorimeter segmentation:

$$[0, 0.2),\ [0.2, 0.4),\ \ldots\ [1.3, 1.37)$$

$$[1.52, 1.6),\ \ldots\ [2.2, 2.4)$$

Crack region $1.37 < |\eta| < 1.52$ excluded.

Quality selection

- **Central cell dominance**: cell $(3, 5)$ must carry highest energy fraction
- **Negative energy clamping**: cells with $e_k < 0$ (noise) set to zero

Closure Test: Distortion Model

Goal: validate the correction in a controlled environment where the "truth" is known by construction.

Distortion model — generate pseudo-data from MC:

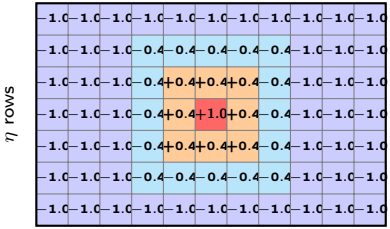
$$e'_k = e_k \cdot \left(1 + \delta \cdot b_k + \varepsilon_k\right)$$

- δ: systematic distortion level (1%, 3%, 5%)
- b_k: fixed bias pattern (narrowing)
- $\varepsilon_k \sim \mathcal{N}(0, \delta/2)$: stochastic noise
- Total cluster energy conserved by rescaling

Quality selection:

- Central cell must have highest energy fraction
- Negative cell energies clamped to zero

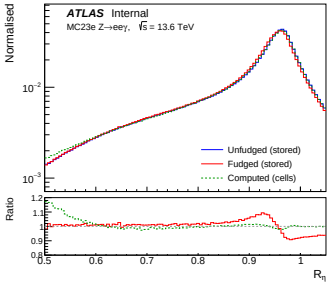
Bias pattern b_k



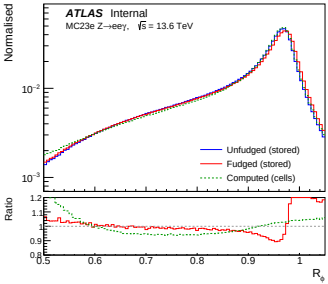
Mimics narrower showers (data-like): enhance core, suppress tails

Validation: Shower Shape Recomputation

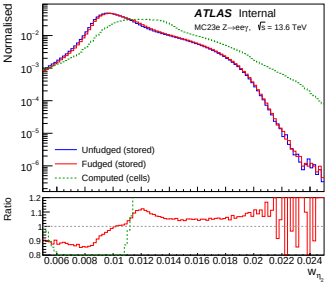
Before running the closure test, verified that shower shapes computed from the 7×11 cell energies **match the stored values** in the ntuples.



R_η — $Z \to ee\gamma$



R_ϕ — $Z \to ee\gamma$



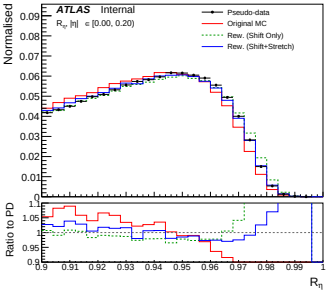
$w_{\eta\,2}$ — $Z \to ee\gamma$

Excellent agreement validates the cell-energy $\to$ shower-shape computation pipeline.

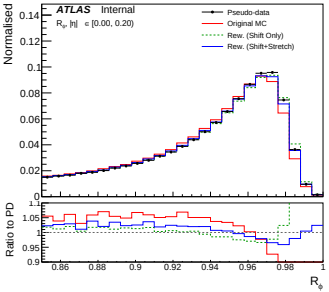
$w_{\eta\,2}$ uses relative cell positions ($\eta_i \in \{0,\ \Delta\eta,\ 2\Delta\eta\}$) — valid since $w_{\eta\,2}$ is translation-invariant.

Closure Results: Central Barrel ($\delta = 5\%$)

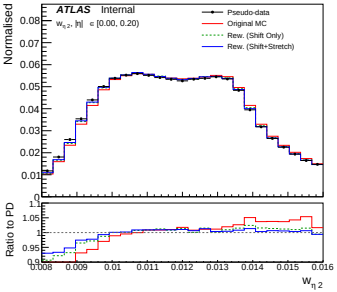
$|\eta| \in [0.00, 0.20)$ — 5% distortion level — full statistics (9.6 M events)



R_η



R_ϕ



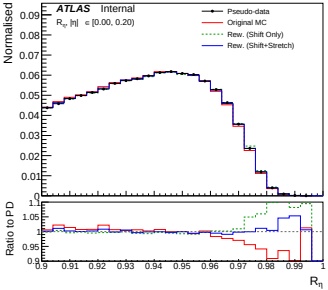
$w_{\eta 2}$

• Pseudo-data — MC (uncorrected) - - Method 1 — Method 2

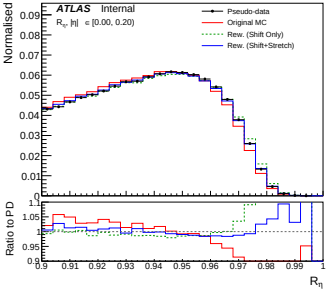
Method 2 brings ratio to within ∼1% of unity across the full range.

Distortion Level Comparison (R_η, Central Barrel)

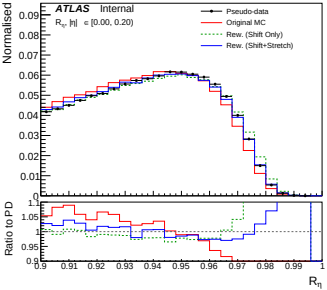
$|\eta| \in [0.00, 0.20)$ — comparing 1%, 3%, and 5% distortion levels



$\delta = 1\%$



$\delta = 3\%$



$\delta = 5\%$

At 1%: distributions nearly indistinguishable. At 5%: discrepancy clearly visible; Method 2 superiority most evident.

Correction performance **scales well** with distortion magnitude.

χ^2/n_{df} Summary ($\delta = 5\%$)

	R_η			R_ϕ			$w_{\eta\,2}$		
$\lvert\eta\rvert$ bin	MC	M1	**M2**	MC	M1	**M2**	MC	M1	**M2**
[0.0, 0.2)	0.0059	0.0046	**0.0005**	0.0052	0.0014	**0.0006**	0.0014	0.0005	**0.0003**
[0.4, 0.6)	0.0063	0.0037	**0.0005**	0.0054	0.0015	**0.0006**	0.0018	0.0007	**0.0004**
[0.8, 1.0)	0.0079	0.0028	**0.0006**	0.0052	0.0021	**0.0003**	0.0022	0.0009	**0.0006**
[1.3, 1.37)	0.0085	0.0026	**0.0008**	0.0051	0.0026	**0.0006**	0.0023	0.0010	**0.0006**
[1.52, 1.6)	0.0078	0.0062	**0.0004**	0.0022	0.0014	**0.0002**	0.0011	0.0016	**0.0007**
[1.8, 2.0)	0.0139	0.0033	**0.0010**	0.0087	0.0038	**0.0004**	0.0032	0.0008	**0.0007**
[2.2, 2.4)	0.0192	0.0035	**0.0010**	0.0118	0.0034	**0.0005**	0.0040	0.0010	**0.0008**

Method 1 (flat shift):

- 50–80% χ^2 reduction

Method 2 (TProfile):

- **90–95% χ^2 reduction**
- Consistent across all $\lvert\eta\rvert$ bins and all 3 variables

Conclusions & Next Steps

Conclusions:

- Cell-energy reweighting method successfully **validated** for Run 3 MC (MC23e) via closure tests
- Tested at 3 distortion levels (1%, 3%, 5%) across 13 populated $|\eta|$ bins
- **Method 2** (energy-dependent TProfile) achieves **90–95%** χ^2/n_{df} reduction — consistently outperforms flat shift (Method 1: 50–80%)
- Correction is **uniform** across all $|\eta|$ bins and all 3 shower shape variables (R_η, R_ϕ, $w_{\eta 2}$)

Next steps:

1. **Data–MC comparison** — apply corrections to real Run 3 data
   - DAOD_EGAM3 ntuples produced (data24 run 473235 + MC23e DSID 700770)
   - Analysis pipeline implemented, pending execution
2. **Conversion-type binning** (derive separate corrections for unconverted / converted photons)
3. **Extend to Layers 1 and 3** (requires new NTupleMaker branches for L1/L3 cell energies)
4. Study impact on **photon ID efficiency** (tight working point)

Backup

Key Adaptations from Reference Method

Differences from ATL-COM-PHYS-2021-640 (Run 2 implementation):

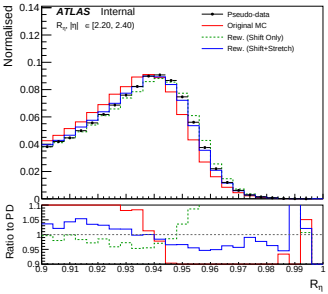
Aspect	Reference (Run 2)	This work (MC23e)
Input format	Matched data–MC pairs	Unmatched MC
Pre-selection	Tight ID + isolation	Central cell + energy clamping
$w_{\eta 2}$ coords	Absolute cell η	Relative positions ($\Delta\eta = 0.025$)
Conv. binning	Inc / UnConv / Conv	Inclusive only[†]
Precision	`float`	`double`

- Central cell cut removes mis-centred/pathological clusters
- Negative energy clamping prevents unphysical fractions ($f_k > 1$ or $f_k < 0$)
- Relative η valid: $w_{\eta 2}$ is a second central moment (translation-invariant)

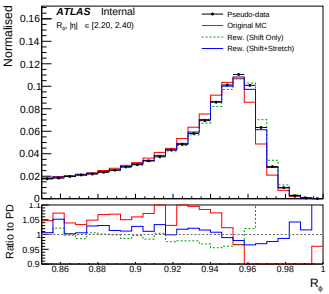
[†] Conversion-type binning planned for data application

Closure Results: Forward Endcap ($\delta = 5\%$)

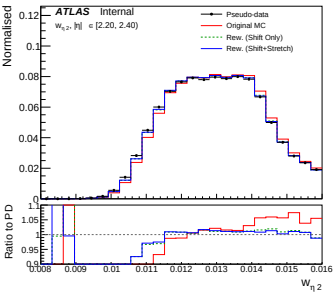
$|\eta| \in [2.20, 2.40)$ — demonstrates method works at high pseudorapidity



R_η



R_ϕ



$w_{\eta\,2}$

Uncorrected χ^2/n_{df} is larger at forward $|\eta|$ (different detector geometry), but Method 2 still achieves $\sim$20$\times$ reduction.

Full χ^2/n_{df}: 5% Distortion

	R_η			R_ϕ			$w_{\eta\,2}$		
$\lvert\eta\rvert$ bin	MC	M1	M2	MC	M1	M2	MC	M1	M2
0	0.0059	0.0046	0.0005	0.0052	0.0014	0.0006	0.0014	0.0005	0.0003
1	0.0064	0.0037	0.0005	0.0053	0.0014	0.0006	0.0017	0.0007	0.0004
2	0.0063	0.0037	0.0005	0.0054	0.0015	0.0006	0.0018	0.0007	0.0004
3	0.0062	0.0035	0.0004	0.0051	0.0014	0.0005	0.0018	0.0008	0.0004
4	0.0079	0.0028	0.0006	0.0052	0.0021	0.0003	0.0022	0.0009	0.0006
5	0.0077	0.0027	0.0005	0.0050	0.0016	0.0003	0.0022	0.0010	0.0006
6	0.0074	0.0027	0.0005	0.0050	0.0016	0.0004	0.0022	0.0011	0.0007
7	0.0085	0.0026	0.0008	0.0051	0.0026	0.0006	0.0023	0.0010	0.0006
9	0.0078	0.0062	0.0004	0.0022	0.0014	0.0002	0.0011	0.0016	0.0007
10	0.0090	0.0038	0.0004	0.0049	0.0019	0.0003	0.0019	0.0011	0.0007
11	0.0139	0.0033	0.0010	0.0087	0.0038	0.0004	0.0032	0.0008	0.0007
12	0.0169	0.0032	0.0010	0.0106	0.0034	0.0005	0.0036	0.0008	0.0007
13	0.0192	0.0035	0.0010	0.0118	0.0034	0.0005	0.0040	0.0010	0.0008

Full χ^2/n_{df}: 3% Distortion

	R_η			R_ϕ			$w_{\eta\,2}$		
$\lvert\eta\rvert$ bin	MC	M1	M2	MC	M1	M2	MC	M1	M2
0	0.0022	0.0016	0.0002	0.0019	0.0005	0.0002	0.0006	0.0002	0.0001
1	0.0023	0.0013	0.0002	0.0020	0.0005	0.0002	0.0006	0.0002	0.0001
2	0.0024	0.0013	0.0002	0.0020	0.0005	0.0002	0.0007	0.0002	0.0001
3	0.0023	0.0013	0.0001	0.0019	0.0005	0.0002	0.0007	0.0003	0.0002
4	0.0029	0.0010	0.0002	0.0020	0.0007	0.0001	0.0008	0.0004	0.0002
5	0.0029	0.0010	0.0002	0.0019	0.0006	0.0001	0.0008	0.0004	0.0002
6	0.0027	0.0010	0.0002	0.0019	0.0006	0.0002	0.0008	0.0004	0.0002
7	0.0035	0.0010	0.0003	0.0021	0.0010	0.0004	0.0009	0.0005	0.0003
9	0.0030	0.0022	0.0002	0.0008	0.0005	0.0001	0.0004	0.0006	0.0003
10	0.0034	0.0014	0.0002	0.0018	0.0007	0.0001	0.0007	0.0004	0.0002
11	0.0052	0.0012	0.0004	0.0032	0.0014	0.0002	0.0012	0.0003	0.0003
12	0.0064	0.0011	0.0004	0.0040	0.0012	0.0002	0.0013	0.0003	0.0002
13	0.0072	0.0013	0.0004	0.0044	0.0012	0.0002	0.0015	0.0003	0.0003

Full χ^2/n_{df}: 1% Distortion

	R_η			R_ϕ			$w_{\eta\,2}$		
$\lvert\eta\rvert$ bin	MC	M1	M2	MC	M1	M2	MC	M1	M2
0	0.0003	0.0002	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
1	0.0003	0.0001	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
2	0.0003	0.0002	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
3	0.0003	0.0002	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
4	0.0004	0.0001	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
5	0.0003	0.0001	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
6	0.0003	0.0001	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
7	0.0005	0.0002	0.0001	0.0003	0.0002	0.0000	0.0001	0.0001	0.0001
9	0.0004	0.0003	0.0000	0.0001	0.0001	0.0000	0.0001	0.0001	0.0000
10	0.0004	0.0002	0.0000	0.0002	0.0001	0.0000	0.0001	0.0000	0.0000
11	0.0006	0.0001	0.0000	0.0004	0.0001	0.0000	0.0001	0.0000	0.0000
12	0.0007	0.0001	0.0000	0.0005	0.0002	0.0000	0.0002	0.0000	0.0000
13	0.0009	0.0002	0.0001	0.0006	0.0002	0.0000	0.0002	0.0000	0.0000

Distortion Model Details

Per-cell distortion:

$$e'_k = e_k \cdot \left(1 + \delta \cdot b_k + \varepsilon_k\right)$$

Bias pattern b_k (Chebyshev distance d_k from centre):

d_k	b_k
0 (central cell)	$+1.0$
1 (first ring)	$+0.4$
2 (second ring)	-0.4
≥ 3 (outer)	-1.0

Energy conservation:

$$e'_k \leftarrow e'_k \cdot \frac{E_{\text{tot}}}{\sum_j e'_j}$$

Quality selection:

- Negative energies $\rightarrow$ clamped to 0
- Central cell must be dominant
- $\sim$81% events pass ($\sim$7.8 M / 9.6 M)

Stochastic noise: $\varepsilon_k \sim \mathcal{N}(0, \delta/2)$ per event per cell — represents irreducible fluctuations. Fixed random seed ensures full reproducibility.

Reference: ATL-COM-PHYS-2021-640

[Figures to be extracted from ATL-COM-PHYS-2021-640.pdf]

F. Caetano et al., "Photon shower shape corrections using cell-level reweighting," ATL-COM-PHYS-2021-640 (2021).

Sections 5.1 (flat shift) and 5.2 (TProfile) — the reference implementation for Run 2 data.